Math 261-001
Quiz 9, November 6

Name	CSU ID #

Choose and answer **exactly two** of the four problems. If you attempt more than two, only the first two will be graded. Write your final answers in the boxes provided. Only answers in the boxes will be graded. You may use the surrounding space for scratch work. Each problem is worth the same number of points.

1. For each item state the **type of integral** (line or surface) and the **vector field being integrated** (curl or original) after applying Stokes' theorem. Do not compute derivatives or evaluate either integral.

   (a) Let $\vec{F} = (y, z, x)$, and let H be the outward-oriented upper hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$. The requested integral is
   $$\iint_H (\operatorname{curl} \vec{F}) \cdot \vec{N}\, d\sigma.$$

   (b) Let $\vec{G} = (z, x, y)$, and let T be the triangle traversed in the order
   $$(2, 0, 0) \longrightarrow (0, 2, 0) \longrightarrow (0, 0, 2) \longrightarrow (2, 0, 0).$$
   The requested integral is $\oint_T \vec{G} \cdot \vec{T}\, ds$.

(a)

(b)

2. Let S be the portion of the paraboloid $z = x^2 + y^2$ for $1 \leq z \leq 4$ with normal pointing outward, away from the z-axis. Give parametrizations of its boundary curves when viewed from above. [Hint: Watch out for the induced orientation].

Outer boundary

Inner boundary

3. Find the flux of $\vec{F} = (y, x, y + z)$ through the portion of $2x + y + z = 2$ in the first octant, oriented **upward**: $\iint_S \vec{F} \cdot \vec{N}\, d\sigma$.

4. Let

$$\vec{F} = (z - x^{2026},\ x - y^{2026},\ y - z^{2026}),$$

and let T be the triangle traversed in the order

$$(2, 0, 0) \longrightarrow (0, 2, 0) \longrightarrow (0, 0, 2) \longrightarrow (2, 0, 0).$$

Use Stokes' theorem to evaluate $\oint_T \vec{F} \cdot \vec{T}\, ds$.